

ChromaIntel LC-MS/MS MVP Dashboard

Dataset size, model comparison, parameter significance, and development roadmap

15052

Master rows

15052

Model rows

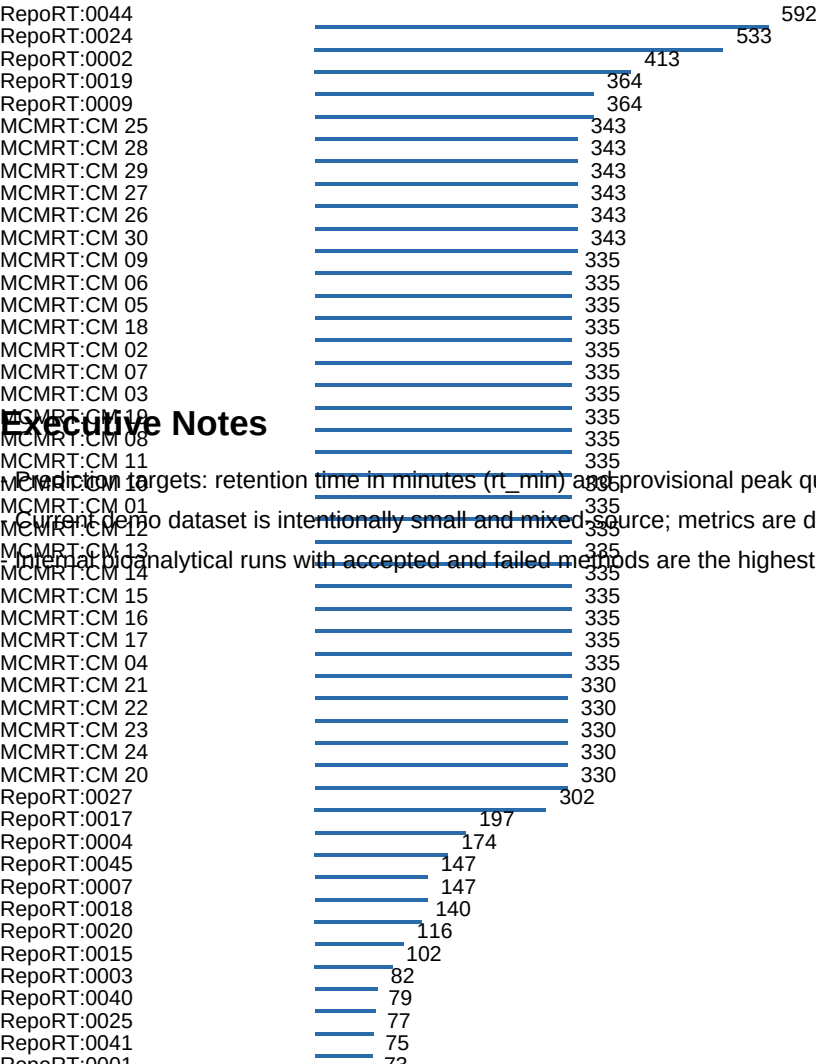
3945

Compounds

75

Sources

Rows by source



Executive Notes

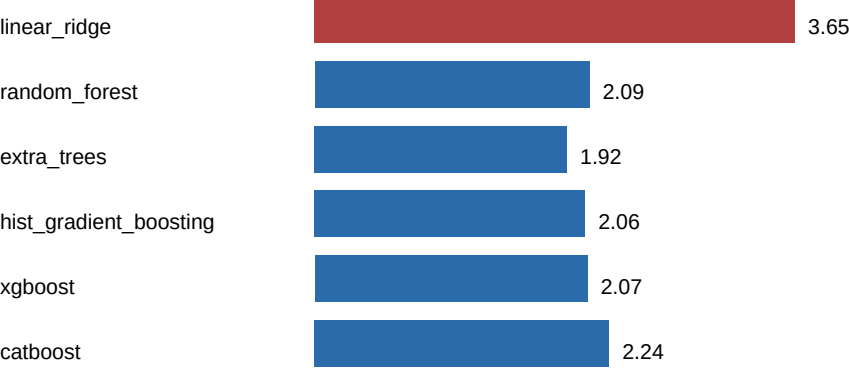
Production targets: retention time in minutes (rt_min) and provisional peak quality score (quality_score).
Current demo dataset is intentionally small and mixed source; metrics are diagnostic, not production evidence.
Internal bioanalytical runs with accepted and failed methods are the highest-value next data source.

Highest missingness fraction

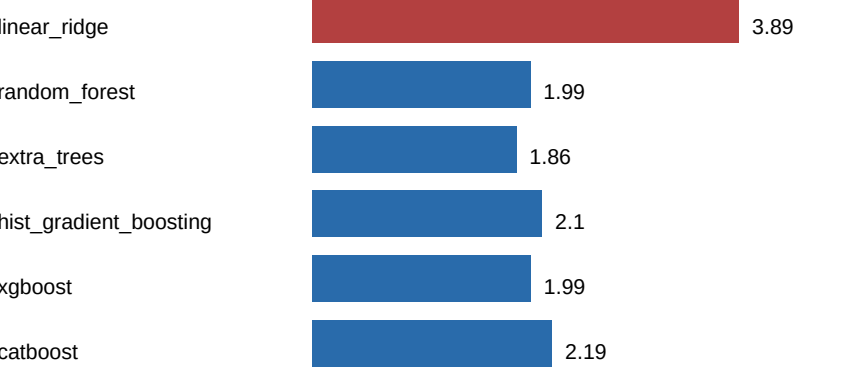


Model Comparison

RT validation MAE (lower is better)



RT test MAE (lower is better)



RT model	Val MAE	Test MAE	Test R2
linear_ridge	3.653	3.891	0.737
random_forest	2.090	1.989	0.904
extra_trees	1.917	1.861	0.920
hist_gradient_boosting	2.056	2.095	0.904
xgboost	2.074	1.991	0.910
catboost	2.238	2.186	0.903

Quality model	Val MAE	Test MAE	Test R2
linear_ridge	0.000	0.000	0.850
random_forest	0.000	0.000	0.938
extra_trees	0.000	0.000	0.948
hist_gradient_boosting	0.000	0.000	0.690
xgboost	0.000	0.000	0.908
catboost	0.000	0.000	0.912

Selected RT model: extra_trees; selected quality model: random_forest

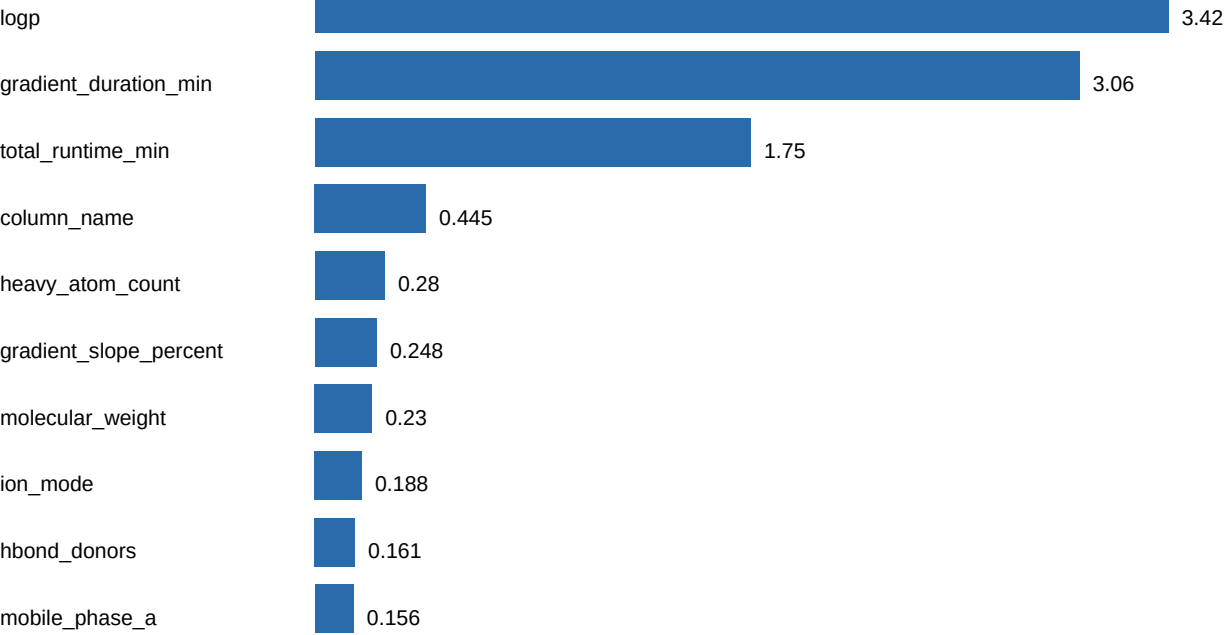
Uncertainty proxy: split_conformal_abs_residual_q90 q90=4.517 min

Source-wise RT error

Source	N	MAE	Bias
RepoRT:0039	14	12.001	0.588
RepoRT:0022	1	11.966	-11.966
MCMRT:CM 16	73	5.856	-1.414
RepoRT:0036	10	4.869	-2.674
RepoRT:0037	7	4.116	-0.273
RepoRT:0002	88	3.414	0.168
MCMRT:CM 15	73	3.376	-0.555
MCMRT:CM 14	73	3.372	-0.696
MCMRT:CM 13	73	3.371	-0.624
RepoRT:0028	13	3.308	-0.913
RepoRT:0032	14	3.218	0.046
MCMRT:CM 12	73	3.181	-0.664

Parameter Significance And Validation Detail

Top RT permutation importance



Feature	Group	Imp.	Signal
logp	compound_descript	3.419	positive
gradient_duration_mi	lc_numeric	3.063	positive
total_runtime_min	lc_numeric	1.746	positive
column_name	lc_categorical	0.445	positive
heavy_atom_count	compound_descript	0.280	positive
gradient_slope_perce	lc_numeric	0.248	positive
molecular_weight	compound_descript	0.230	positive
ion_mode	ms_setting	0.188	positive
hbond_donors	compound_descript	0.161	positive
mobile_phase_a	lc_categorical	0.156	positive

Held-out prediction diagnostics

Test rows: 3247 | RT MAE: 1.861 min | RT RMSE: 3.173 min | AD flags: 0

Compound	Source	Actual RT	Pred RT	Abs Error
sophorose	RepoRT:0002	1.640	53.724	52.084
Salicylic acid	RepoRT:0039	81.600	39.720	41.880
Pelargonidin	RepoRT:0039	41.990	81.524	39.534
Clenbuterol	MCMRT:CM 16	22.870	48.984	26.114
Destruxin B	MCMRT:CM 16	57.770	36.709	21.061
Gliotoxin	MCMRT:CM 16	41.680	21.924	19.756
emodin	RepoRT:0002	46.405	28.130	18.275
Gallic acid	RepoRT:0039	16.540	33.757	17.217
Alternariol	MCMRT:CM 16	49.870	33.018	16.852
HT-2 Toxin	MCMRT:CM 16	48.910	32.327	16.583
Propazine	MCMRT:CM 16	51.760	35.618	16.142

Roadmap

0-2 weeks: data foundation

- Import internal BE-style historical runs with accepted, failed, low-intensity, poor-resolution, and carryover cases.
- Expand reviewed public RT samples and preserve source/license/missingness provenance.
- Add import preview summaries for invalid rows, duplicate runs, and missing MS/LC fields.

2-4 weeks: stronger validation and models

- Move from random split to grouped/source holdout by InChIKey, source, and method family.
- Evaluate Morgan fingerprints, ExtraTrees, CatBoost/LightGBM/XGBoost when enough rows exist.
- Persist applicability-domain ranges and penalize out-of-domain recommendations.

1-2 months: recommendation quality

- Add score decomposition: RT fit, quality, runtime, confidence, AD penalty.
- Make LC search space configurable by lab-approved columns, solvents, pH, flow, temperature, and runtime.
- Collect outcome feedback from recommended methods for active learning.

2-4 months: SOTA research track

- Pretrain or benchmark graph/Transformer structure encoders on METLIN/RepoRT/MCMRT-scale RT data.
- Use retention-order objectives where absolute RT does not transfer cleanly between methods.
- Add calibrated conformal intervals by matrix, source, column chemistry, and instrument platform.

Decision gate: do not claim production readiness until internal data volume, grouped validation, and lab acceptance criteria are in place.